PRACTICE QUESTIONS.

By Pin2

MTH632

Lecture No. 3

Modules (8 – 9)

Q1: Verify that $|z|^2 = z\bar{z}$ for $z = -3 + 4i$.

Solution:

$z = -3 + 4i$, so $\bar{z} = -3 + 4i = -3 - 4i$

$z\bar{z} = (-3 + 4i)(-3 - 4i)$

$= (-3)(-3) + (-3)(-4i) + (4i)(-3) + (4i)(-4i)$

$= 9 + 12i - 12i - 16\, i^2$

$= 9 - 16\,(-1)$

$z\bar{z} = 25$

Now $|z| = \sqrt{(-3)^2+(4)^2} = \sqrt{25} = 5$

$|z|^2 = 5^2 = 25$

Hence $|z|^2 = z\bar{z}$

Q2: Verify that $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ for $z_1 = 2 + i$, $z_2 = -3 + 9i$.

Solution:

$\bar{z}_1 + \bar{z}_2 = (2+i) + (-3+9i)$

$= (2-3) + (i+9i)$

$= -1 + 10i$

$= -1 - 10i$

$$\bar{z}_1 + \bar{z}_2 = \overline{(2+i)} + \overline{(-3+9i)}$$

$$= 2 - i + (-3-9i)$$

$$= -1 - 10i$$

$$\text{Hence} \quad \bar{z}_1 + \bar{z}_2 = \overline{z_1 + z_2}$$

Q3: Show that $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$ and $\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}$.

Solution:

$$z = x + yi, \quad \text{and} \quad \bar{z} = x - yi$$

$$\mathrm{Re}(z) = x, \qquad \mathrm{Im}(z) = y$$

$$\text{Now } z + \bar{z} = (x + yi) + (x - yi)$$

$$= (x + x) + (yi - yi)$$

$$= 2x + 0i$$

$$z + \bar{z} = 2x$$

$$\frac{z + \bar{z}}{2} = x$$

$$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}$$

$$\text{Now } z - \bar{z} = (x + yi) - (x - yi)$$

$$= (x - x) + (yi + yi)$$

$$= 0 + 2yi$$

$$z - \bar{z} = 2yi$$

$$\frac{z - \bar{z}}{2i} = y$$

$$\mathrm{Im}(z) = \frac{z - \bar{z}}{2i}$$

Q4: Show that $\mathrm{Re}(iz) = -\mathrm{Im}(z)$ and $\mathrm{Im}(iz) = \mathrm{Re}(z)$.

Solution:

$$\text{Since } \mathrm{Re}(z) = \frac{z + \bar{z}}{2}$$

$$\text{So } \mathrm{Re}(iz) = \frac{iz + \overline{iz}}{2}$$

$$= \frac{iz - i\bar{z}}{2}$$

$$= i\left(\frac{\bar{z} - z}{2}\right)$$

$$= i.i\left(\frac{\bar{z} - z}{2i}\right)$$

$$\mathrm{Re}(iz) = -\mathrm{Im}(z)$$

$$\text{Since } \mathrm{Im}(z) = \frac{z - \bar{z}}{2i}$$

$$\text{So } \mathrm{Im}(iz) = \frac{iz - \overline{iz}}{2i}$$

$$= \frac{iz + i\bar{z}}{2i}$$

$$= i\left(\frac{z + \bar{z}}{2i}\right)$$

$$= \left(\frac{z + \bar{z}}{2}\right)$$

$$\mathrm{Im}(iz) = \mathrm{Re}(z)$$

Q5: Verify that the triangle inequality $|z_1 + z_2| \le |z_1| + |z_2|$ for $z_1 = -8 + i$, $z_2 = -3 + 8i$.

Solution:

$z_1 = -8 + i, \quad |z_1| = \sqrt{(-8)^2+1^2} = \sqrt{65} = 8.06$

$z_2 = -3 + 8i, \quad |z_2| = \sqrt{(-3)^2+8^2} = \sqrt{73} = 8.5$

$z_1 + z_2 = (-8 + i) + (-3 + 8i) = -11 + 9i$

$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{202} = 14.12$

Clearly $|z_1+z_2| < |z_1| + |z_2|$

14.12 < 8.06+8.5

14.12 < 16.56

Q6: Find the Polar representation of the complex numbers $z = -7 + 5i$.

Solution:

We represent $z = (x, y)$ in (r, θ) in polar form.

For $z = x + iy, \quad r = |z| = \sqrt{(-7)^2+5^2} = \sqrt{74}$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{-7}\right) = -\tan^{-1}\left(\frac{5}{7}\right)$$

$z = r(\cos\theta + i\sin\theta)$

$$z = \sqrt{74}\left(\cos\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)+i\sin\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)\right)$$

Q7: Find the Polar representation of the complex numbers $z = (0, 11) = 11i$.

Solution:

We represent $z = (x, y)$ in (r, θ) in polar form.

For $z = x + iy, \quad r = |z| = \sqrt{(0)^2+11^2} = \sqrt{121} = 11$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{11}{0}\right) = \infty, \quad \text{so } \theta = \frac{\pi}{4}$$

$z = r(\cos\theta + i\sin\theta)$

$$z = 11\left(\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right)$$